

Jayateja Alugolu

Toronto, ON, CA · alugolu.jayateja@gmail.com · 506-897-5428 · linkedin.com/in/jayateja-alugolu · github.com/jayateja-a

SUMMARY

Software Engineer with 5 years of experience in Java, Spring Boot, React, REST APIs, and microservices, building and supporting applications across fintech, healthcare, telecommunications, and enterprise environments. Hands-on experience with **AWS, Azure, Kubernetes, Docker, Terraform, CI/CD, databases, and distributed systems**, with a focus on **scalable, secure, and reliable applications**.

EDUCATION

University of New Brunswick, Canada

Jan 2024 – Aug 2025

Master of Computer Science | GPA: 3.8

- ◆ Relevant Coursework: Foundations of Privacy, Network Security, Big Data Systems, Natural Language Processing, Advanced Software Process, Cloud Info Management, Data Analytics, Digital Forensics, Business Analytics.

PROFESSIONAL EXPERIENCE

Software Developer — LightX Innovations

Oct 2025 – Feb 2026

Fredericton, Canada (Remote) | Contract Full time

- ◆ Developed and maintained a **Java Spring Boot Backend** for a mobile ophthalmology application enabling clinicians to capture and upload eye images, securely associate them with patient records, and store them in a **FHIR-compliant** and **DICOM-standardized healthcare data platform**.
- ◆ Built and maintained a **multi-tenant cloud platform** using **Docker, Kubernetes, Azure Kubernetes Service (AKS), and Helm charts**, implementing **tenant isolation** across production workloads, **documented deployment processes, architecture, and operational workflows**.
- ◆ Built **TypeScript-based data migration pipelines** and **DICOM image processing workflows** to transform and standardize legacy clinical data, using **FFmpeg video compression** and **H.265 encoding techniques** to optimize media processing and storage, reducing data migration time by **50%**.
- ◆ Developed secure Azure cloud infrastructure using **Terraform**, integrating **Azure Blob Storage for healthcare images** and **PostgreSQL for FHIR-compliant metadata** to support scalable and interoperable clinical data management.
- ◆ Used **AI-assisted development tools** such as **GitHub Copilot** and **Cursor** to speed up **Java/Spring Boot development**, refactor code, write unit tests, troubleshoot issues, and improve overall development efficiency while maintaining code quality.
- ◆ Implemented **CI/CD pipelines** using **Azure DevOps** and **GitHub Actions**, automating build, testing, and deployment workflows and improving release efficiency by **40%**.

Full Stack Security Developer Intern — Canadian Institute for Cybersecurity

Jan 2025 – Apr 2025

Fredericton, Canada

- ◆ Developed and enhanced a **Social Threat Intelligence platform** using a **Laravel (PHP) frontend**, integrating backend **FastAPI (Python) services** to enable real-time monitoring and interactive visualization of large-scale security datasets, including social media derived threat intelligence, sentiment analysis, dashboards, and geospatial (map-based) insights.
- ◆ Designed and implemented **high-performance RESTful APIs** using **FastAPI**, migrating from Flask-based services to **asynchronous, scalable microservices** with modular architecture, request validation, and optimized data access patterns.
- ◆ Built and optimized a **search and analytics backend** using **Elasticsearch**, ingesting and indexing large volumes of third-party social data, enabling fast querying and filtering of threat-related content across multiple categories.
- ◆ Increased system reliability by achieving **90%+ unit test coverage** using **pytest** and **FastAPI TestClient**, significantly reducing regression issues during production releases.
- ◆ Implemented **Kafka-based event streaming** for event-driven microservices, enabling asynchronous communication between services and improving the scalability of security data processing workflows.
- ◆ Configured **Prometheus and Grafana** for application monitoring and performance diagnostics, helping identify service bottlenecks and investigate production issues.
- ◆ Containerized microservices using **Docker** and deployed them to **Kubernetes**, supporting scalable and repeatable cloud-native deployments.

Software Engineer Intern — Quber Technologies Inc.

May 2024 – Aug 2024

Fredericton, Canada

- ◆ Developed **Java Spring Boot microservices and REST APIs** for employee savings and financial transactions, handling **transaction orchestration, idempotency, concurrent requests, database consistency, and failure scenarios**, implemented **Spring transaction management, validation, centralized exception handling, JWT authentication** across distributed services.
- ◆ Built **React and Redux-based fintech application** for employee onboarding and financial services, implementing **reusable**

components, Redux state management, REST API integration, form validation, routing, asynchronous data handling, and responsive UI design.

- ◆ Created and maintained **technical documentation and KT materials** covering application setup, deployment processes, troubleshooting procedures, and system workflows.
- ◆ Implemented **Spring Security with OAuth2/JWT-based authentication and authorization**, providing secure access across **multi-tenant financial applications** and enforcing role-based access to protected resources.
- ◆ Improved backend performance by **reducing API latency by 30%** through **MySQL query optimization, Redis caching, and service-level tuning**, improving responsiveness for transaction-related operations.
- ◆ Deployed and supported production workloads on **AWS**, using **EC2, S3, ALB, DynamoDB, and IAM** to support highly available applications and secure service communication.
- ◆ Worked through **staging-to-production CI/CD workflows**, troubleshooting deployment issues and supporting reliable releases across development and production environments.

Senior Systems Engineer — Tata Consultancy Services

Sep 2022 – Dec 2023 (1+ years)

Hyderabad, India

- ◆ Developed enterprise-grade backend services for the **British Telecom Global Services Portal** using **Java, Spring Boot, and Angular**, supporting scalable service integrations across distributed enterprise systems.
- ◆ Built and maintained **RESTful APIs using Java Spring Boot**, implementing **request validation, pagination, rate limiting, and exception handling** to improve reliability and consistency across service integrations.
- ◆ Developed and executed **unit and integration tests using JUnit and Mockito**, validating service-layer logic, REST endpoints, repository interactions, and integration scenarios before production releases.
- ◆ Improved performance of high-traffic transactional systems by optimizing **Oracle SQL queries and Hibernate ORM mappings**, reducing query execution time by **55%** and resolving database bottlenecks affecting more than **10,000 concurrent users** during peak load.
- ◆ Participated in **Agile development, production troubleshooting, architecture discussions, and code reviews**, while mentoring junior developers and coordinating technical decisions across distributed teams.

Systems Engineer — Infosys Limited

Jul 2019 – Sep 2022 (3+ years)

Hyderabad, India

- ◆ Worked on the **modernization of a legacy workforce management platform for BNSF Railway**, developing **Java Spring Boot microservices and RESTful APIs**, supporting application development and production issues, and maintaining **SOAP integrations** with existing enterprise systems.
- ◆ Integrated **legacy mainframe services with Spring Boot microservices**, enabling reliable communication with **DB2-based enterprise systems** while gradually replacing legacy service orchestration components.
- ◆ Implemented dynamic frontend features using **JavaScript, jQuery**, and **AJAX-based workflows** to enable seamless interaction with backend **REST APIs**.
- ◆ Designed and optimized **MySQL queries and stored procedures** supporting complex transactional workflows and improving database performance for business-critical operations.
- ◆ Troubleshot production issues across **Java services, database integrations, APIs, and legacy systems**, reducing recurring incidents by **25%** through automation, monitoring improvements, and CI/CD enhancements.

PROJECTS

Serverless Threat Detection — AWS

Feb 2025

- ◆ Developed an AI-driven intrusion detection system using **Random Forest & Isolation Forest** for anomaly detection, built an online analysis pipeline with **Glue & SageMaker** for automated forensic analysis, Stored real-time logs in **DynamoDB** and triggered workflows using **AWS Lambda & EventBridge**.

BankAPI — Modular Financial Backend

Oct 2024

- ◆ Developed a **Java Spring Boot** banking API supporting account creation, fund transfers, and transaction history using a layered architecture with **DTOs, services, and repositories** for scalability and maintainability, Implemented **validation, error handling, and rate limiting**, and delivered full API documentation with Swagger to enable seamless integration and testing, Designed transaction workflows with a focus on **idempotent API operations, data consistency, and reliable request processing** for duplicate or retried requests.

SKILLS

C++ | Java | Python | JavaScript | TypeScript | HTML/CSS | React | Redux | Angular | Spring Boot | Hibernate | Django | Flask | FastAPI | Node.js | Microservices | Distributed Systems | MySQL | PostgreSQL | DynamoDB | Redis | Kafka | AWS | Azure | Docker | Kubernetes | Terraform | CI/CD | Jenkins | GitHub Actions | Azure DevOps | Git | REST APIs | SOAP | JSON | GraphQL | Postman | Swagger | ETL Pipelines | Elasticsearch | Linux | Agile | Scrum | Data Structures and Algorithms | Teamwork |